03_sass_tailwindcss_en

Laboratory 03

Task 1

SASS - System Design Tokens + Padel Tiles
SASS skills: Variables, Nesting, Mixins, Loops, @extend, Partials

Objective

A statistics dashboard (e.g. for padel) with a full design token system (colours, spacing, shadows, breakpoints) – build a complete statistics dashboard (e.g. for padel) from scratch using a modular SCSS architecture.

STEP BY STEP:

1. Build a system of design tokens in separate partials files (_colors.scss , _spacing.scss , _shadows.scss , _breakpoints.scss) - minimum 15 zmiennych w mapach Sass
2. Write 4 parametric mixins (card layout, hover effects, breakpoints, gradienty)
3. Design 6 statistics tiles (wins, matches, ELO, average points, tournaments, win rate) arranged in a grid
4. Implement an @for loop to generate unique background gradients for each tile
5. Use @extend to create a button system (primary, secondary, danger)
6. Make it fully responsive – desktop (3x2), tablet (2 columns), mobile (1 column)
7. Add dark mode using the `html.dark` selector with `@if/@else` logic
8. Use Sass nesting for all hover/focus/active states

FINAL RESULT:

An interactive dashboard featuring smooth animations, a professional colour scheme and full responsiveness.

📁 File structure:

```
sass/
├── _variables.scss     # design tokens
```

```
├── _mixins.scss        # hover, shadows
├── _tiles.scss         # component
└── main.scss           # import + root
```

Requirements:

```
•   10+ SCSS variables (e.g. $padel-primary, $spacing-xs, $shadow-md)
•   3 mixins:  @mixin hover-lift ,  @mixin card ,  @mixin breakpoint($size)
•   @for loop: 6 tiles with different statistics
•   @extend dla shared button styles
•   Nesting for hover/focus states
```

HTML template:

```html
<div class="dashboard">
  <div class="tile wins"><span>47</span> Victories</div>
  <div class="tile matches"><span>124</span> Matches</div>
  <!-- 4 kolejne -->
</div>
```

Estimated SCSS code (~80 lines):

```scss
// _variables.scss
$padel-primary: #10b981;
$spacing-xs: 0.5rem;
$shadow-md: 0 4px 12px rgba(0,0,0,0.15);

// main.scss
@use 'variables';
@use 'mixins';

.tile {
  @include mixins.card;
  @for $i from 1 through 6 {
    &:nth-child(#{$i}) { background: hsl(#{160 + $i*20}, 70%, 90%); }
```

```
  }
}
```

Estimated time: 60 mins

Task 2

TailwindCSS - User Profile with Animations
Tailwind skills: Utilities, Custom config, Plugins, JIT, Arbitrary values

Objective

A complete player profile (like LinkedIn + Strava) with custom Tailwind configuration – Create a modern padel player profile (Strava/LinkedIn style) with full TailwindCSS customisation.

STEP BY STEP:

1. Create a tailwind.config.js file with an extended theme: padel colours (#10b981), custom spacing, shadows, and the 'padel-sm' breakpoint
2. Build a 3-section layout:
   - Hero section: Avatar with a gradient border, name with gradient text, ELO badge with pulse, basic statistics
   - Stats grid: 6 tiles (3x2 desktop) with hover glow effects
   - Activity timeline: Last 10 matches (date, opponent, result, emoji)
3. Implement a custom plugin adding the class bg-gradient-padel
4. Use arbitrary values (w-17.3rem , ring-2.5 , translate-y--4px)
5. Add group-hover variants to the statistics tabs
6. Create animations (pulse for ELO, bounce for buttons, slide-in for the timeline)
7. Full responsiveness + dark mode (class strategy)
8. Timeline with scroll-snap and hover states

FINAL RESULT:

A sports profile featuring glassmorphism effects, animations, dark mode and flawless responsiveness.

Requirements:

```
✅ Layout: Hero + Stats Grid + Activity Timeline
✅ Custom config: kolory padel (#10b981), spacing[2.6], shadows
✅ Animacje: spin, pulse, slide-in (Tailwind animate-* + custom)
✅ Responsive: mobile→desktop (sm:, md:, lg:)
✅ Dark mode: class strategy (html.dark)
✅ Arbitrary values: w-[17.3rem], ring-2.5
✅ Plugin: custom 'padel-gradient'
```

HTML template:

```html
<!DOCTYPE html>
<html class="dark">
<head>
  <script src="https://cdn.tailwindcss.com"></script>
  <script>
    tailwind.config = { theme: { extend: { colors: { padel: '#10b981' }}}}
  </script>
</head>
<body class="bg-gradient-to-br from-slate-50 to-blue-50 dark:from-slate-900">
  <div class="max-w-4xl mx-auto p-8">
    <!— Profil hero, stats, timeline —>
  </div>
</body>
</html>
```

Expected result:

```html
<div class="group/card hover:scale-[1.02] hover:shadow-2xl hover:shadow-padel/50
            bg-gradient-to-r from-padel to-emerald-500/80
            p-8 rounded-3xl animate-pulse-once">
  <h3 class="font-black text-3xl text-white drop-shadow-md">ELO: 1567</h3>
</div>
```

Estimated time: 60 mins